Assignment 2: Model Exploration & Results

1. Model Selection & Justification

In this study, the problem is treated as a binary classification task because the target variable, Potability, has two possible outcomes: potable or not potable. The dataset includes 3,276 water samples and uses 9 numerical features—pH, hardness, solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity—to predict whether water is safe to drink. The class distribution, with 1,998 non-potable and 1,278 potable samples, also makes classification an appropriate approach.

The first model selected for this study is Logistic Regression. This model was chosen because it is simple, effective for binary classification, and easy to interpret. Since all variables in the dataset are numerical and the missing values were handled during preprocessing, Logistic Regression provides a suitable baseline model. In the code, missing values in pH, Sulfate, and Trihalomethanes were filled with the mean, which made the dataset ready for modeling. Another reason for choosing this model is that it helps explain the general relationship between the input variables and water potability, which is useful for understanding the data more clearly.

The second model selected is Random Forest Classifier. This model is more suitable for capturing complex and non-linear relationships between variables. Based on the correlation matrix and other visualizations in the analysis, there do not appear to be very strong linear relationships between individual features and potability. This suggests that water potability may depend on the combined effect of several features rather than a single variable. Random Forest is a good choice in this case because it can model feature interactions, handle complex patterns, and perform well with noisy environmental data. It can also provide feature importance scores, which help identify the most influential water quality indicators.

These model choices also support the project's objective related to SDG 6: Clean Water and Sanitation. The presentation highlighted that traditional laboratory testing is often slow, expensive, and difficult to scale. In contrast, AI-based models can support faster, more scalable, and more proactive water quality assessment. Logistic Regression offers a simple and interpretable starting point, while Random Forest provides stronger predictive power. Together, these models create a balanced approach by combining interpretability and performance, which can contribute to improving access to safe and clean water.

In conclusion, Logistic Regression and Random Forest Classifier are both appropriate choices for this dataset and problem. Logistic Regression serves as a clear baseline model, while Random Forest has the potential to achieve better performance by capturing more complex patterns in the data. Therefore, using both models provides a solid foundation for the modeling stage of this project.

2. Data Preparation for Modeling

Before model training, the dataset was prepared to make it suitable for machine learning analysis. First, missing values in the pH**,** Sulfate, and Trihalomethanes variables were handled using mean imputation in order to avoid information loss and keep the dataset complete for modeling. After that, the dataset was divided into input features **(x)** and the target variable **(y)**, where **Potability** was used as the prediction target. Since this is a classification problem, the data was then split into training and test sets so that model performance could be evaluated on unseen data. To preserve the class distribution in both subsets, a stratified train-test split was applied. In addition, because the dataset contains numerical variables measured on different scales, standardization was applied where appropriate, especially for models such as Logistic Regression that are sensitive to feature scale. No major feature engineering was performed, since the dataset already includes a compact set of directly relevant water quality indicators. These preprocessing steps helped ensure that the data was clean, consistent, and suitable for the modeling stage.

3. Model Training & Evaluation

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Logistic Regression
Accuracy: 0.6097560975609756
Precision: 0.0
Recall: 0.0
F1-score: 0.0
ROC-AUC: 0.5484179687499999

Classification Report:
              precision    recall  f1-score   support

           0       0.61      1.00      0.76       400
           1       0.00      0.00      0.00       256

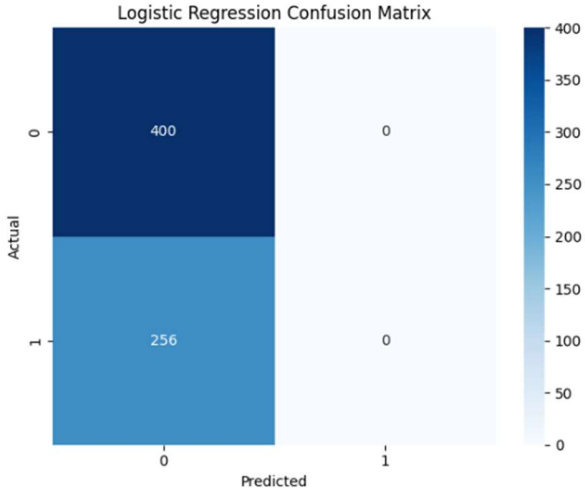
    accuracy                           0.61       656
   macro avg       0.30      0.50      0.38       656
weighted avg       0.37      0.61      0.46       656

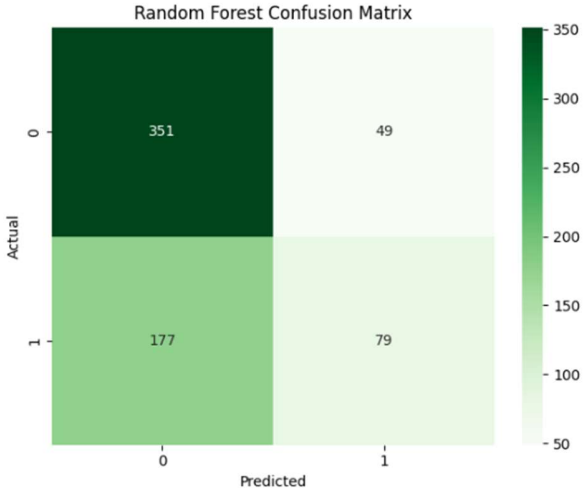

Random Forest
Accuracy: 0.6554878048780488
Precision: 0.6171875
Recall: 0.30859375
F1-score: 0.4114583333333333
ROC-AUC: 0.6548388671875001

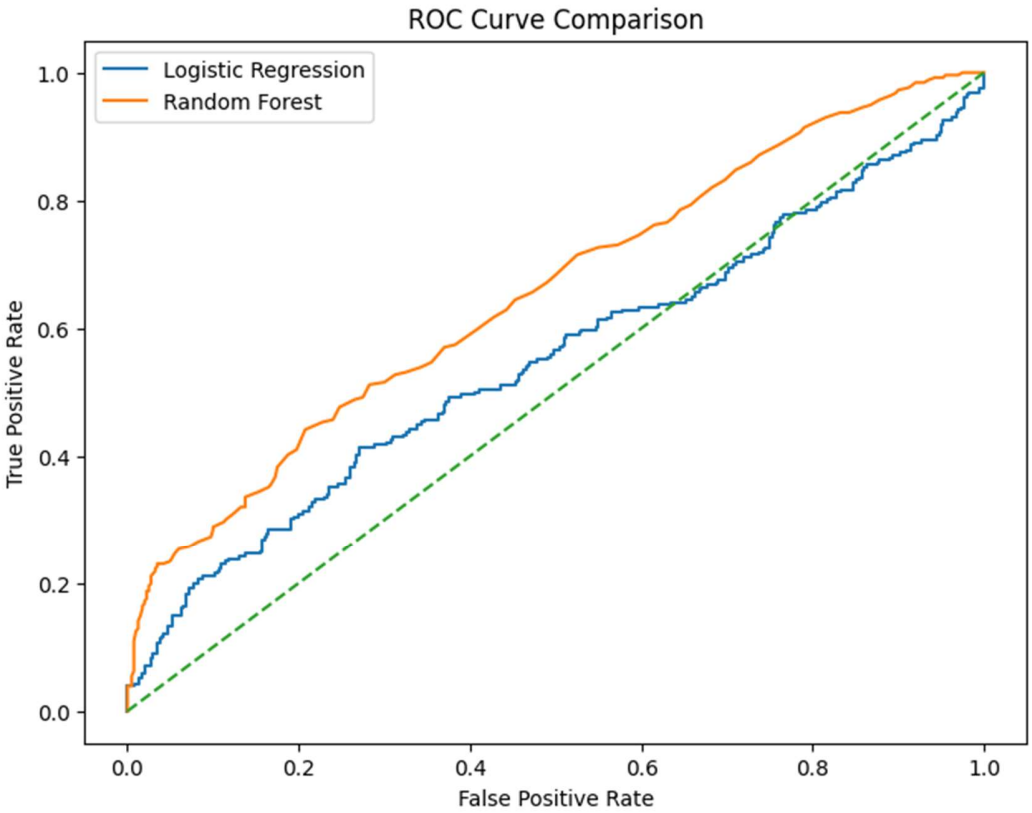
Classification Report:
              precision    recall  f1-score   support

           0       0.66      0.88      0.76       400
           1       0.62      0.31      0.41       256

    accuracy                           0.66       656
   macro avg       0.64      0.59      0.58       656
weighted avg       0.65      0.66      0.62       656
```


Logistic Regression Confusion Matrix
400
0
256
0
Actual
Predicted
0
1
400
350
300
250
200
150
100
50
0


Random Forest Confusion Matrix
351
49
177
79
Actual
Predicted
0
1
350
300
250
200
150
100
50


ROC Curve Comparison
Logistic Regression
Random Forest
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0

4. Interpretation of Results

4.1. Interpretation of Model Outputs in Relation to Project Objectives

The primary objective of this study is to predict water potability using machine learning models in alignment with Sustainable Development Goal 6 (Clean Water and Sanitation). In this context, the performance of Logistic Regression and Random Forest models was evaluated.

The results indicate that the Logistic Regression model achieved an accuracy of approximately 0.61. However, its performance on the positive class (potable water) was extremely poor, with precision and recall values equal to 0.00. This indicates that the model failed to correctly identify any potable water samples. The ROC-AUC score of approximately 0.55 further confirms that the model performs only slightly better than random guessing.

In contrast, the Random Forest model demonstrated improved performance, achieving an accuracy of approximately 0.65 and an ROC-AUC score of around 0.65. Unlike Logistic Regression, Random Forest was able to identify both classes, with a precision of 0.62 and recall of 0.31 for potable water. Although the recall value remains relatively low, the model shows a significantly better ability to capture non-linear relationships within the data.

From an SDG perspective, correctly identifying potable water is critical. Therefore, despite having similar accuracy levels, the Random Forest model is more suitable due to its ability to detect potable water samples, even if imperfectly.

4.2. Key Insights and Unexpected Findings

One of the most significant findings of this study is the impact of class imbalance on model performance. The dataset contains a higher number of non-potable water samples compared to potable ones. As a result, the Logistic Regression model is biased toward the majority class and predicts almost all samples as non-potable.

Another important insight is that accuracy alone is not a sufficient metric for evaluating model performance in this context. Despite achieving an accuracy above 60%, Logistic Regression completely fails to identify potable water samples. Therefore, metrics such as recall, F1-score, and ROC-AUC provide a more comprehensive evaluation.

Furthermore, although Random Forest performs better, its recall for potable water remains relatively low (0.31), indicating that a significant portion of potable water samples are still misclassified. This is a critical issue, especially in real-world applications where missing a potable water source may lead to incorrect decision-making.

4.3. Limitations of the Study

Several limitations can be identified in the current approach. First, the dataset suffers from class imbalance, which negatively affects the learning process of the models. Second, no advanced feature engineering techniques were applied, limiting the models' ability to capture complex relationships between variables.

Additionally, hyperparameter tuning was not performed, particularly for the Random Forest model, which may have restricted its full potential. The use of mean imputation for handling missing values may also have reduced data variability and introduced bias into the dataset.

Finally, the relatively low recall for potable water indicates that the models are not yet reliable for real-world applications where accurate detection is essential.

4.4. Recommendations for Improvement

To improve model performance, several steps can be taken. First, techniques such as SMOTE or class weighting can be applied to address class imbalance. Second, hyperparameter tuning should be performed to optimize model performance.

Moreover, more advanced models such as Gradient Boosting (e.g., XGBoost or LightGBM) could be explored. Feature engineering techniques, including interaction terms and domain-specific transformations, may also enhance predictive power.

Finally, future studies should prioritize evaluation metrics such as recall and F1-score, particularly for the potable water class, to ensure alignment with SDG objectives.